LOCAL FOOD WASTAGE MANAGEMENT SYSTEM

Data Analytics & Insights Report

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1. Executive Summary

This report outlines the development of a **Food Waste Management Dashboard** that has been created using the programming languages of Streamlit and MySQL to perform analysis on food listings, claims, and food distribution.

The main aim of the project is to minimize food waste through the provision of useful information for effective communication between the food provider and receiver.

2. Project Overview

Project Description

The Dashboard serves as an integrated tool to track the availability of food, its distribution, and claims made.

Problem Statement

- High amount of food wastage due to non-coordinated efforts
- Ineffective Distribution System
- No real-time tracking mechanism

Objectives

- Analysis of availability and requirement of food
- Inefficiencies in food distribution
- Reduction of food wastage

3. Dataset Description

Source of Data

The source of data used for the project is from different datasets in MySQL, including:

- **food_listings_data**

- **providers_data**
- **receivers_data**
- **claims_data**

❖ Dataset Structure

1. Providers Dataset

Column Name	Data Type	Description
Provider_ID	Integer	Unique identifier for each provider
Name	String	Name of the food provider (restaurant, grocery store, etc.)
Type	String	Category of provider (Restaurant, Grocery Store, Supermarket)
Address	String	Physical address of the provider
City	String	City where the provider is located
Contact	String	Contact details (phone/email)

2. Receivers Dataset

Column Name	Data Type	Description
Receiver_ID	Integer	Unique identifier for each receiver
Name	String	Name of the receiver (individual/organization)
Type	String	Category (NGO, Community Center, Individual)
City	String	City where the receiver is located
Contact	String	Contact details

3. Food Listings Dataset

Column Name	Data Type	Description
Food_ID	Integer	Unique identifier for each food item
Food_Name	String	Name of the food item
Quantity	Integer	Quantity available for distribution
Expiry_Date	Date	Expiry date of the food item
Provider_ID	Integer	Reference to provider
Provider_Type	String	Type of provider
Location	String	City where food is available
Food_Type	String	Category (Vegetarian, Non-Vegetarian, Vegan)
Meal_Type	String	Meal type (Breakfast, Lunch, Dinner, Snacks)

4. Claims Dataset

Column Name	Data Type	Description
Claim_ID	Integer	Unique identifier for each claim
Food_ID	Integer	Reference to food item
Receiver_ID	Integer	Reference to receiver
Status	String	Claim status (Pending, Completed, Cancelled)
Timestamp	Datetime	Date & time of claim

Key Properties

- Location: Location of food
- Type of Food: Type of food
- Quantity: Quantity of food
- Expiry: Validity of food
- Status of Claim: Status of request

4. Methodology

Data Cleaning

- Missing data removed
- Date/numeric formatting applied
- Data standardization

Data Transformation

- Aggregation using SQL
- Calculating key performance indicators
- Grouping/filtering data

Tools used

- Python
- Streamlit
- MySQL
- Pandas
- Plotly

5. Dashboard features

- Interactivity through KPI cards
- Dynamic filters (city, food type, date range)
- Data visualization (line chart, bar chart, pie chart)
- Map visualization
- Advanced SQL analysis (18 queries)

    - ✓ The total number of SQL queries used in the dashboard is 18. From those, 13 SQL queries were mentioned in the given problem statement, whereas 5 other queries were individually created by us to get further insight into the data, such as food wastage, performance of providers, and unclaimed food.

6. Key Metrics (KPIs)

- Listings: Denotes number of food listings
- Quantity: Number of food units in listings
- Average quantity: Average number of food items per listing
- Claims: Number of food claims
- Top location: City making maximum contribution

7. Data Analysis and insights

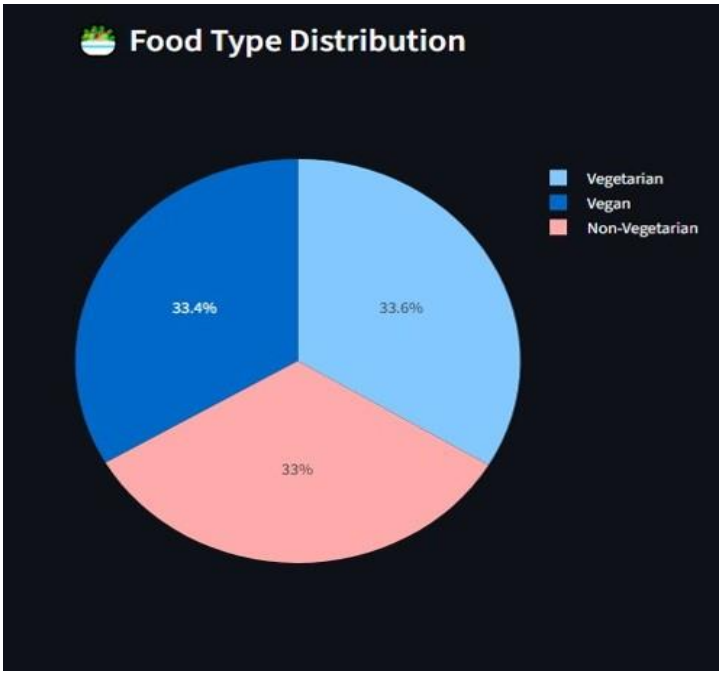
Trend analysis

- Food availability differs among various dates
- Peaks denote high surplus of food
- Few troughs suggest underutilization



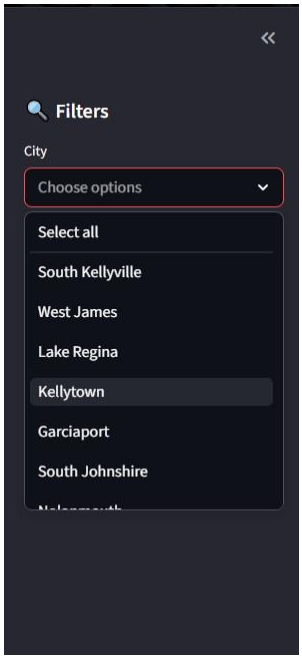
Category Analysis

- Specific food categories make up most of the listings
- Low diversity of foods



Location Analysis

- Major cities provide the bulk of the food
- Certain locations experience low availability of food



8. Visual Analysis

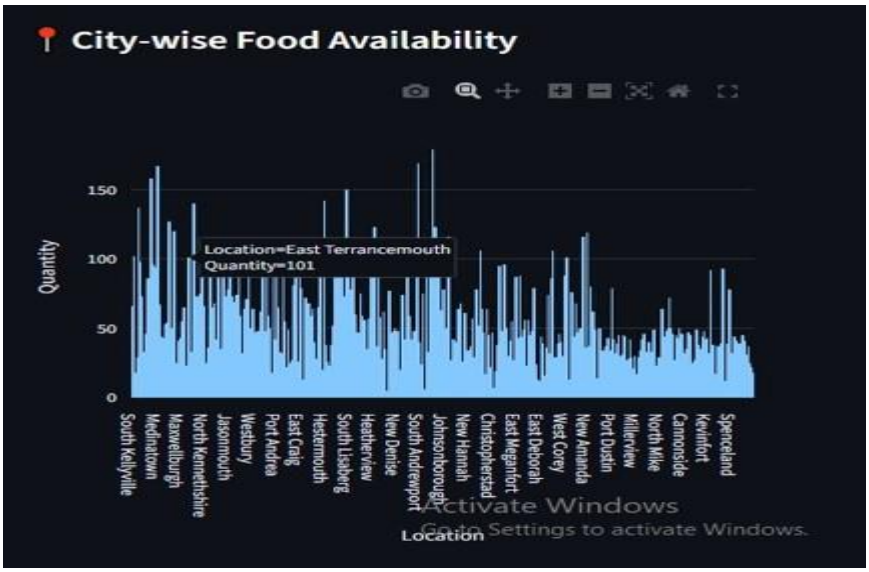
Graph (Trend)

- Shows how the amount of food changes over time.
- Allows determining when food availability is at its peak.



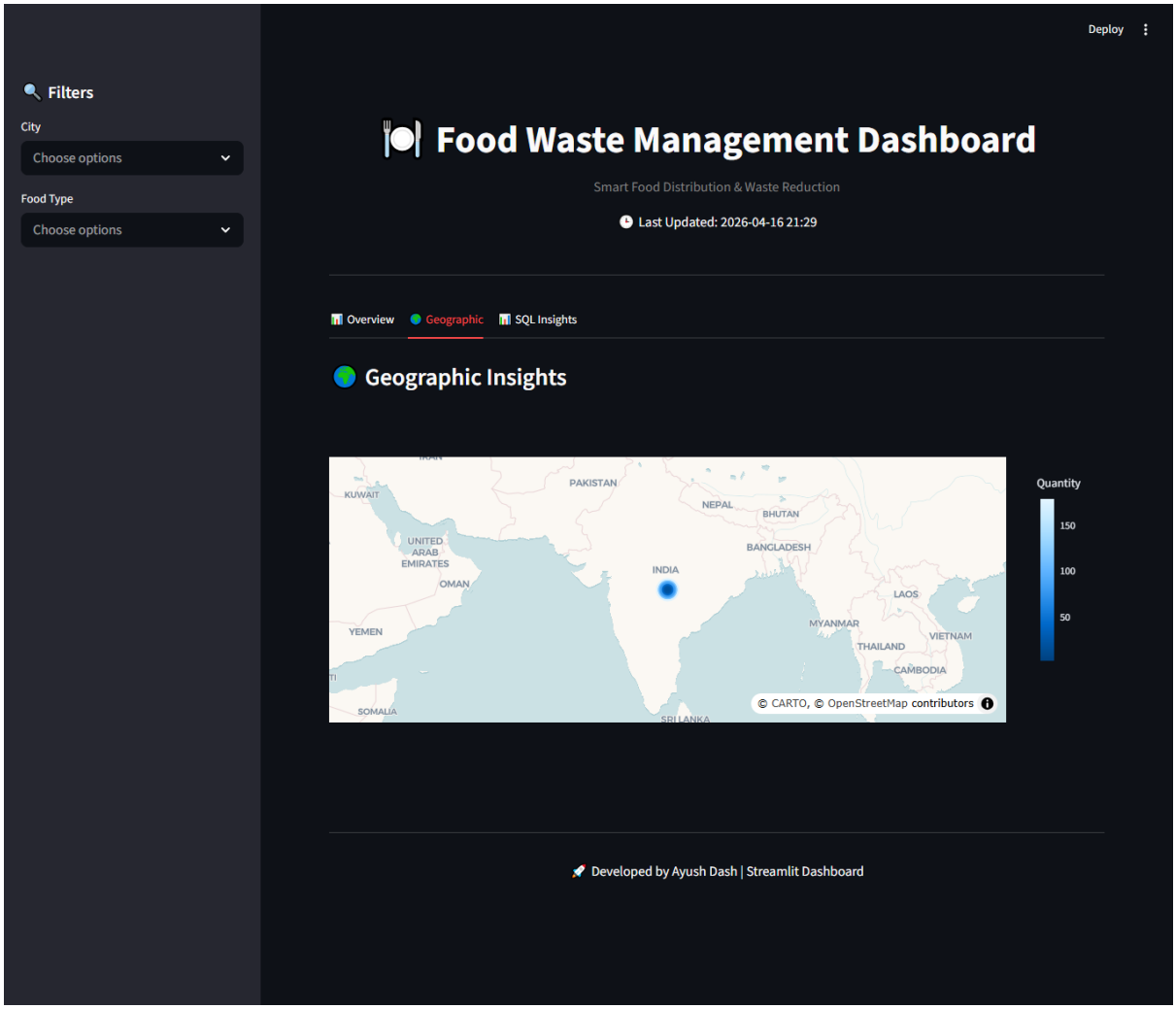
Graph (City Comparison)

- Shows how many foods are provided by each city.
- Determines which cities contribute the most.



Geo-Mapping

- Shows how the food availability depends on geography.
- Larger circles represent more abundant food.



9. Key Findings

- Food availability is inconsistent over time
- A small number of cities contribute the majority of food
- Significant portion of food remains unclaimed
- Distribution is uneven across categories and locations
- System lacks real-time balancing of supply and demand

10. Conclusion

The Food Waste Management Dashboard provides a centralized and data-driven approach to monitor food availability, distribution, and claims.

It helps identify inefficiencies, reduce food wastage, and improve coordination between providers and receivers.

11. Recommendations

- Coordinate effectively with receivers and providers.
- Develop real-time alerts about availability of foods.
- Transfer excess food to places where demand exceeds supply.
- Raise awareness among receivers.
- Streamline claim process.